2) $g = 7 \quad p = 23$

For A, $x = 3$

$a = g^x \bmod p$

$= 7^3 \bmod 23$

$= 343 \bmod 23$

$= 21$

For B, $y = 5$

$b = g^y \bmod p$

$= 7^5 \bmod 23$

$= 16807 \bmod 23$

$= 17$

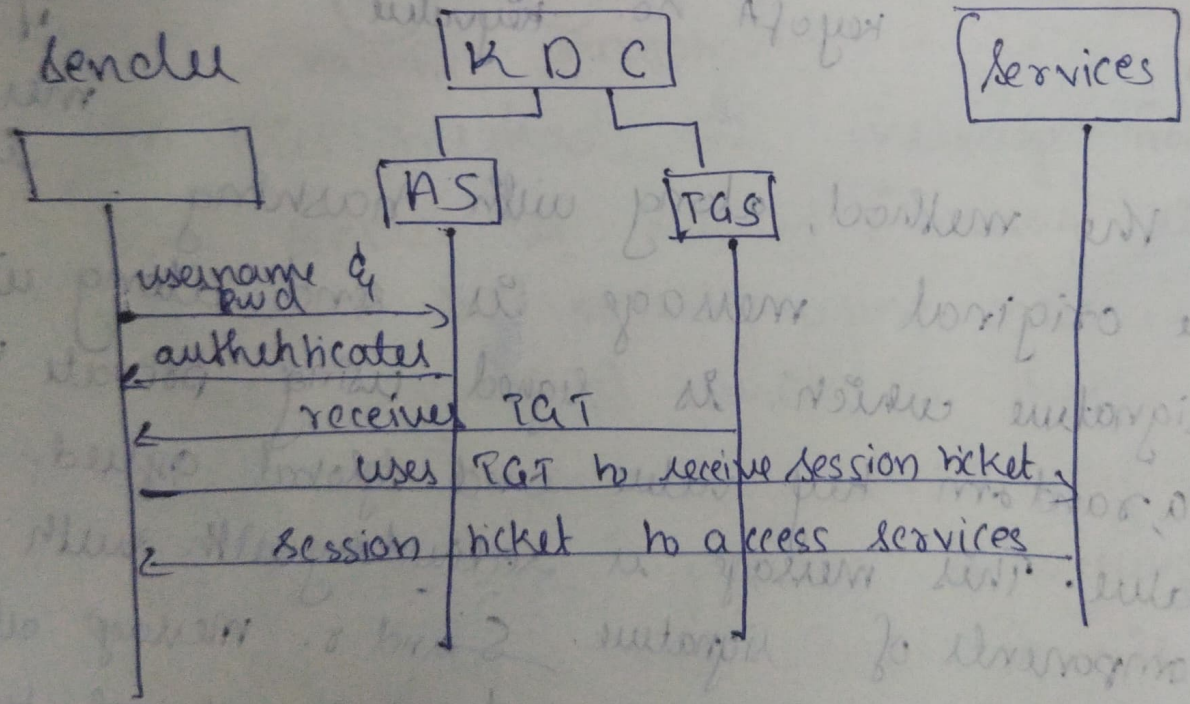
exchange 21 & 17 b/w A & B

$a_{new} = 17^3 \bmod 23 = 14$

$b_{new} = 21^5 \bmod 23 = 14$

$\therefore$ Shared key $= 14$

5) Sender



KDC - Key distribution Centre

AS - Authentication Server

TGS - Ticket Generation System

TGT - Ticket Generation Ticket

Kerberos is a third party authentication system which removes the need for repeated Authentication via passwords. The Key distribution Centre is the most important component of Kerberos which has the key distribution database which deals with authentication and ticket generation. ~~Kerber~~ KDC has 2 servers

- Authentication Server: This server authenticates the user in the kerberos environment
- TGS - Ticket Distribution Server which is responsible for issuing tickets that grant access to the system.

<u>Working</u>.

a) Login

The user logs into the Kerberos interface using his/her password and username and this authenticated and verified by the Authentication server

b) The user is then directed to the ~~authe~~ Ticket generation server that issues tickets

that grant access to the environment

c) The user uses these tickets to access each service for which it applies for permission using TGT and receives session tokens

d) These session tokens along with TGT can be used to access the various services provided.

Advantages

- Kerberos provide a secure means of authentication without having to send the actual password over the network
- It can be used as one-time authentication means

Disadvantages

- If the KDC fails, the entire system fails (single point of failure).